Prediction Markets Are Fisher Markets

Batch Auction Clearing via Eisenberg-Gale Duality

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Summary. We prove that prediction market batch auctions with budget-constrained market makers are Fisher markets (Theorem 5). Replacing linear market-maker welfare with logarithmic utility — reflecting the Kelly-criterion sizing that real institutional MMs use — transforms the budget-constrained clearing problem into a convex program with a unique solution, solvable in polynomial time. Budget constraints vanish from the formulation entirely, absorbed into the Eisenberg-Gale objective. Prices, fills, and budget allocation emerge simultaneously from a single convex optimization.

1. Introduction

The main result of this paper is that prediction market batch auctions are Fisher markets.

A prediction market exchange runs *batch auctions*: orders accumulate, then clear simultaneously at uniform prices. The clearing problem is an allocation problem — find the fills and prices that maximize welfare subject to balance constraints. Without budget constraints, this is a standard Linear Program.

Market makers introduce the main difficulty. An ordinary participant submits a single order: "buy 100 shares of Yes at \$0.60." The capital required (\$60) is known at submission and locked upfront — no budget constraint needed. A market maker, by contrast, posts *hundreds* of orders across *dozens* of markets simultaneously. The total capital consumed depends on which orders fill and at what prices — both determined by the auction. The MM deposits a finite balance B_k and the exchange must ensure total spending does not exceed it. This budget constraint is *bilinear*: capital consumed depends on the clearing price, and the clearing price depends on which fills happen. This single constraint makes the feasible set non-convex.

We prove three things, in order:

1. **The framework (§2).** LP batch clearing and Hanson's LMSR are the same mathematical object at different temperatures, connected by Fenchel duality. This is the foundation for everything that follows. It is also where we prove that clearing prices are unique when budgets are absent — a fact that makes the budget obstacle precise.
2. **The obstacle (§3).** Adding budget constraints to the risk-neutral model makes the feasible set non-convex (bilinear constraints, indefinite Hessian). Standard convex optimization no longer applies, no polynomial-time algorithm is known, and uniqueness of clearing prices is an open question. The structural cause: linear welfare with hard budget caps is an internally inconsistent economic model.
3. **The resolution (§§4–5).** Replacing linear MM welfare with Kelly-criterion utility ($B_k \ln U_k$) transforms the problem into a strictly convex Eisenberg-Gale program. Budget constraints are absorbed into the objective — they do not appear as constraints at all. The program has a unique optimum, unique clearing prices, and is polynomial-time solvable. This is the Fisher market isomorphism.

The modeling "trick" is not logarithmic utility — it is the linear assumption. When we fix it, the pathology fixes itself.

2. Foundations: Batch Clearing and LMSR

This section builds the mathematical framework for the main result. Readers familiar with cost-function market makers (Abernethy et al. 2013) will recognize these results in a new framing.

2.1. Minting Cost and the LP

Consider a prediction market with K mutually exclusive outcomes. In a batch auction, N orders arrive. Each order i has a limit price L_i, a maximum quantity $\overline{Q}_i$, a side (buy or sell), and a target market $m(i)$. Some orders belong to market makers; we write MM_k for the set of orders belonging to MM k.

The *net demand* — excess buy over sell volume — for each outcome is:

$$D_k = \sum_{i \in \text{ buy}(k)} q_i - \sum_{j \in \text{ sell}(k)} q_j$$

where $q_i \in \left[0, \overline{Q}_i\right]$ is the fill quantity of order i.

To clear the market, the exchange *mints* complete sets: one share of every outcome at cost \$1 (exactly one resolves to \$1 at settlement, so this is fairly priced). To supply D_k net shares of each outcome, exactly $\max_k D_k$ mints are needed — this covers the highest-demand outcome, with surplus shares of other outcomes left over. The minting cost is therefore $V(\boldsymbol{D}) = \max_k D_k$, and the welfare-maximizing clearing solves:

$$P: \quad \max_{\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]} \sum_i w_i q_i - \max_k D_k(\boldsymbol{q})$$

where w_i is the welfare coefficient of order i ($+L_i$ for buyers, $-L_i$ for sellers) and $\overline{\boldsymbol{Q}} = \left(\overline{Q}_1, ..., \overline{Q}_N\right)$ is the vector of max fill quantities. Since V is convex and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$, the objective is concave — this is a convex optimization problem. Introducing an explicit minting variable $M \geq D_k$ for each k, this is equivalent to the Linear Program:

$$\max_{\boldsymbol{q}, M} \sum_i w_i q_i - M \quad \text{s.t.} \quad D_k(\boldsymbol{q}) \leq M \quad \forall k, \quad \boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$$

Its dual variables are clearing prices.

2.2. Fenchel Duality: Prices from Conjugates

The constraint that clearing prices must be probabilities ($\sum p_k = 1,\ p_k \geq 0$) is not imposed by fiat — it is the Fenchel conjugate of the minting cost. Minting at \$1 per complete set *encodes* the probability axiom.

Theorem 1 (Minting–Simplex Duality). *The Fenchel conjugate of* $V(\boldsymbol{D}) = \max_k D_k$ *is the indicator function of the probability simplex:*

$$V^*(\boldsymbol{p}) = \delta_\Delta(\boldsymbol{p}) = \begin{cases} 0 & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

where $\Delta = \left\{\boldsymbol{p} \geq 0 : \sum_k p_k = 1\right\}$.

Proof. For $\boldsymbol{p} \in \Delta$: convexity gives $\sum p_k D_k \leq \max_k D_k$, so the supremum is 0 (attained at $\boldsymbol{D} = \boldsymbol{0}$). For $\boldsymbol{p} \notin \Delta$: some deviation from the simplex exists, and scaling $\boldsymbol{D}$ in the exploiting direction sends the supremum to $+\infty$. □

The probability axiom $\sum p_k = 1$ is not a modeling choice — it is a no-arbitrage condition. Any deviation from the simplex is an arbitrage opportunity (e.g., if $\sum p_k > 1$, mint at \$1 and sell at $\sum p_k$), and the conjugate enforces this as a hard constraint.

2.3. Entropy Smoothing: LMSR as Soft LP

Hanson's LMSR cost function $C_{b(\boldsymbol{D})} = b \ln \sum \exp(D_k/b)$ is a smooth (C^∞) version of the minting cost, parametrized by a temperature $b > 0$. The structural content is in its Fenchel conjugate: where $V^* = \delta_\Delta$ was a hard constraint (prices must be probabilities), C_b^* is a soft penalty (prices near uniform are cheap, prices near a vertex are expensive).

Theorem 2 (LMSR–Entropy Duality). *The Fenchel conjugate of C_b is negative Shannon entropy on the simplex:*

$$C_b^*(\boldsymbol{p}) = \begin{cases} b \sum_k p_k \ln p_k & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

Proof. We compute $C_b^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \{\sum p_k D_k - b \ln \sum \exp(D_k/b)\}$ by setting the gradient to zero:

$$p_k - \frac{\exp(D_k/b)}{\sum_j \exp\bigl(D_j/b\bigr)} = 0 \quad \Rightarrow \quad p_k = \frac{\exp(D_k/b)}{\sum_j \exp\bigl(D_j/b\bigr)}$$

This is the *softmax* — exactly the LMSR marginal price. Inverting: $D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp\bigl(D_j/b\bigr)$. Substituting back:

$$\langle \boldsymbol{p}, \boldsymbol{D} \rangle = b \sum_k p_k \ln p_k + b \ln Z, \qquad C_b(\boldsymbol{D}) = b \ln Z$$

So $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k$. This is finite only when $\boldsymbol{p} \in \Delta$ (the softmax always yields a probability vector). □

The approximation quality is controlled by b:

Proposition 1 (LSE–Max Sandwich). $\max_k D_k \le C_{b(\boldsymbol{D})} \le \max_k D_k + b \ln K$. *The gap $b \ln K$ is the maximum LMSR subsidy.*

Proof. $\exp(\max_k D_k/b) \le \sum \exp(D_k/b) \le K \exp(\max_k D_k/b)$. Apply $b \ln$. □

The complete picture:

$$\begin{array}{ccc} V = \max_k D_k & \xleftrightarrow{\text{Fenchel}} & V^* = \delta_\Delta \\ \uparrow\ b \to 0 & & \uparrow\ b \to 0 \\ C_b = b \ln \sum \exp(D_k/b) & \xleftrightarrow{\text{Fenchel}} & C_b^* = b \sum p_k \ln p_k \end{array}$$

As $b \to 0$, the smooth cost C_b sharpens to the LP cost V, and the entropy penalty hardens to the simplex indicator.

2.4. The Smoothed Batch Auction and Its KKT Conditions

Replace the minting cost V with C_b in the batch clearing:

$$P_b : \quad \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \quad \sum_i w_i q_i - C_{b(\boldsymbol{D}(\boldsymbol{q}))}$$

Since C_b is convex and smooth, and $\boldsymbol{D}(\boldsymbol{q})$ is linear, P_b is a smooth concave maximization. Its first-order conditions are necessary and sufficient, and the exponentials in the clearing prices come from the minting cost, not the order book.

Theorem 3 (LMSR = Smoothed Batch Clearing). *At the optimum of P_b, the clearing prices are the softmax of net demand:*

$$p_k^* = \frac{\partial C_b}{\partial D_k} = \frac{\exp(D_k^*/b)}{\sum_j \exp\left(D_j^*/b\right)}$$

This is the LMSR marginal price function. By construction, $\sum_k p_k^* = 1$.

The two limits are immediate: as $b \to 0$, $P_b \to P$ (by Berge's theorem on the compact feasible set, with uniform convergence from Proposition 1); as $b \to \infty$, $\exp(D_k/b) \to 1$ for all k, so $p_k \to 1/K$.

Applying KKT with box constraints $q_i \in \left[0, \overline{Q}_i\right]$ yields the familiar clearing rule (for a buy order i on outcome k, the marginal welfare is $L_i - p_k$):

- $q_i = \overline{Q}_i$ (fully filled) $\quad\Leftrightarrow\quad L_i \geq p_k$ *(buyer's limit above price)*
- $q_i = 0$ (unfilled) $\quad\Leftrightarrow\quad L_i \leq p_k$ *(buyer's limit below price)*
- $0 < q_i < \overline{Q}_i$ (marginal) $\quad\Leftrightarrow\quad L_i = p_k$ *(buyer is the price-setter)*

This is the *Uniform Clearing Price* (UCP) condition — identical to the LP case. The entropy smoothing does not alter order-matching logic; it only changes how prices depend on quantities: continuously via softmax rather than discretely via the marginal order.

2.5. Price Uniqueness Without Budgets

Without budget constraints, clearing prices are *always* unique — for any $b > 0$, any order book, unconditionally. This is proved by passing to the Fenchel dual, where the entropy term provides strict convexity. Write $W(\boldsymbol{D})$ for the maximum welfare achievable at a given demand vector $\boldsymbol{D}$ (the inner LP over fills $\boldsymbol{q}$ with $\boldsymbol{D}$ fixed). The primal problem in demand space is $\max_{\boldsymbol{D}}[W(\boldsymbol{D}) - C_b(\boldsymbol{D})]$, where W is concave piecewise-linear. By Fenchel-Rockafellar duality, this is equivalent to minimizing over prices:

$$\min_{\boldsymbol{p} \in \Delta} \left[\underbrace{W^*(\boldsymbol{p})}_{\text{consumer surplus}} + \underbrace{C_b^*(\boldsymbol{p})}_{\text{entropy penalty}} \right]$$

where $W^*(\boldsymbol{p})$ is the total surplus of orders whose limit prices exceed clearing prices, and $C_b^*(\boldsymbol{p}) = b \sum p_k \ln p_k$ is the negative Shannon entropy (Theorem 2).

Theorem 4 (Unconstrained Price Uniqueness). *The Fenchel dual of the unconstrained smoothed problem is strictly convex: W^* is convex (piecewise linear) and C_b^* is strictly convex on the interior of Δ. The clearing prices $\boldsymbol{p}^*$ are therefore unique for any $b > 0$ and any order book, unconditionally.*

Proof. W^* is convex (conjugate of concave W) and C_b^* is strictly convex on the simplex interior: its Hessian is $b \cdot \mathrm{diag}(1/p_k)$, positive definite for $\boldsymbol{p} > 0$. The minimizer lies in the interior because $\nabla C_b^* = b(1 + \ln p_k) \to -\infty$ as any $p_k \to 0^+$, which dominates the finite gradient of W^* and pulls the minimizer away from the boundary. Strict convexity on the interior gives uniqueness. □

3. The Budget Obstacle

When market makers have budget constraints, the clearing problem becomes fundamentally harder. Without budgets, batch clearing is a standard convex program (Theorem 4). With budgets, the capital constraint is bilinear — capital consumed depends on clearing prices, which depend on fills — and standard convex optimization no longer applies.

3.1. The Bilinear Constraint

A market maker k deposits balance B_k and posts orders across multiple markets. The capital consumed by each fill depends on the clearing price:

$$\mathrm{cap}_{k(\boldsymbol{p},\boldsymbol{q})} = \sum_{i \in \mathrm{MM}_k} c_{i\left(p_{m(i)}\right)} \cdot q_i, \quad c_{i(p)} = \begin{cases} p & \text{if BuyYes/SellNo} \\ 1-p & \text{if SellYes/BuyNo} \end{cases}$$

The budget constraint $\mathrm{cap}_k \leq B_k$ is *bilinear*: p is determined by $\boldsymbol{q}$ through the clearing mechanism. The product $c(p(\boldsymbol{q})) \cdot q$ makes the feasible set non-convex in fill space. This single constraint is what separates prediction market clearing from a standard LP.

3.2. Computational Consequences

The bilinear budget constraint has three consequences for computation:

1. The feasible set is non-convex. Define $h_k(\boldsymbol{q}) = \sum_{i \in \mathrm{MM}_k} p_{m(i)}(\boldsymbol{q}) \cdot q_i$ as MM k's capital consumption. We show the Hessian is indefinite via the simplest non-trivial case: an MM with buy orders on two independent binary groups, one order per group, fill quantities q_1, q_2. Since groups are independent, $h = f(q_1) + f(q_2)$ where $f(q) = \sigma(q/b) \cdot q$ and σ is the logistic sigmoid. Differentiating:

$$f''(q) = \frac{\sigma(1-\sigma)}{b}\left[2 + \frac{q(1-2\sigma)}{b}\right]$$

Since $\sigma(1-\sigma) > 0$, the sign depends on $2 + q(1-2\sigma)/b$. At $q = 0$: $f''(0) = 1/(2b) > 0$ (convex). For large q: $\sigma \to 1$ and the bracket becomes $2 - q/b \to -\infty$ (concave). The inflection point $f''(q^*) = 0$ is at $q^* \approx 2.4b$. With fills straddling the threshold — $q_1 < q^*$ and $q_2 > q^*$ — the diagonal Hessian $\mathrm{diag}(f''(q_1), f''(q_2))$ is indefinite. Sublevel sets $\{h \leq B_k\}$ are not guaranteed convex.

2. No standard convex algorithm applies. The unconstrained problem P_b is a smooth concave maximization (solved by interior point in polynomial time). Adding budget constraints $\{h_k \leq B_k\}$ makes the feasible set non-convex. Standard convex optimization — interior point, projected gradient, Frank-Wolfe — requires convex feasible sets. The budget-constrained problem is a bilinear program, for which no polynomial-time algorithm is known in general.

3. The uniqueness question is open. The proof of Theorem 4 (price uniqueness without budgets) uses strict convexity of the Fenchel dual. With budgets, the argument breaks: uniqueness at one price vector requires *cross-price budget feasibility* ($\mathrm{cap}_k(\boldsymbol{p}^2, \boldsymbol{q}^1) \leq B_k$), which the bilinear constraint does not guarantee. Whether clearing prices are nonetheless unique for all LMSR instances is an open question. (In the degenerate case of two identical markets with a symmetric MM, two KKT points exist by symmetry — but this requires exact parameter tuning and does not extend to generic order books.)

Proposition 2 (Computational Obstruction). *The budget-constrained risk-neutral clearing problem is a bilinear program: the feasible set $\{\boldsymbol{q} \in \mathcal{C} : h_k(\boldsymbol{q}) \leq B_k\}$ is non-convex (indefinite constraint Hessian). Standard convex optimization does not apply. By contrast, the risk-averse program P_b^{RA} (Theorem 5) is a standard convex program solvable in polynomial time.*

In practice, exchanges handle budgets by iterative heuristics: solve the LP ignoring budgets, check for violations, adjust, repeat. Such methods have no convergence guarantee and can cycle. The risk-averse program eliminates this entirely — budgets are absorbed into the objective.

3.3. The Expenditure Perspective

A natural idea: change variables to capital expenditures $e_i = c_{i(p)} \cdot q_i$, which linearizes the budget constraint to $\sum_{i \in \mathrm{MM}_k} e_i \le B_k$. In Fisher markets, the Eisenberg-Gale program works precisely because this change of variables produces a convex program. Does it work here?

No. Welfare transforms to $\sum_i w_i \cdot e_i / c_{i(p)}$ — a *rational* function of prices. In Fisher markets (Eisenberg & Gale 1959), the analogous program is convex because agents have *diminishing-returns* utilities: $\sum B_k \ln U_k$ provides curvature. Our MMs have constant marginal returns (linear welfare), providing none. The expenditure substitution linearizes the budget but introduces non-convexity into the objective.

The diagnosis is structural: *linear welfare with hard budget caps* is an internally inconsistent economic model — agents simultaneously risk-neutral (constant marginal value) and risk-averse (capped exposure). The non-convexity is the computational symptom.

The mathematics tells us exactly what is needed: diminishing-returns utility to provide curvature. §4 argues that this is not a mathematical convenience but the independently correct economic model — the computational need and the economic reality coincide. §5 proves the resulting program is convex.

4. Why Market Makers Have Diminishing Returns

The previous section showed that clearing requires diminishing-returns utility to be computationally tractable. This section argues that diminishing returns is also the independently correct economic model for repeat-participation MMs — the mathematical fix and the economic reality are the same object. We give three theoretical arguments and two empirical observations.

4.1. Kelly Criterion Is a Survival Theorem

Breiman (1961) proved that Kelly maximization (log utility) is the *unique* repeated-game strategy that maximizes long-run growth rate a.s., reaches any wealth target in minimum expected time, and dominates any other strategy with probability 1. An MM that sizes linearly faces ruin with probability 1 given enough batches.

A prediction market exchange runs *repeated* batch auctions, compounding returns. The natural objective for a repeat participant is $\max \sum_t \mathbb{E}[\ln(1 + r_t)]$ — exactly the Kelly criterion, exactly log utility per batch. Linear utility maximizes one-shot expected value but minimizes survival across batches. Since FBAs are inherently repeated, log utility is the only consistent objective.

Modeling MMs with log utility is not an assumption but a *selection effect*: the MMs that survive long enough to matter are the ones already using Kelly-like sizing.

4.2. Budget Constraints *Are* Risk Aversion

Why do MMs have budget constraints at all? A risk-neutral agent with positive expected value should bet everything. The existence of $B_k <$ total wealth is itself evidence of risk aversion: the MM limits exposure because it values capital preservation.

Linear welfare with a hard budget is internally inconsistent: "I value each share equally (risk-neutral) but I won't risk more than B_k (risk-averse)." The budget is a crude piecewise-linear approximation of what log utility handles smoothly.

4.3. Empirical Grounding

Order books reveal concave demand. Real MMs do not post one order at one limit price for their entire budget. They post *ladders*: multiple orders at decreasing limit prices with decreasing quantities. Each tranche has lower welfare and smaller size —

diminishing returns, expressed directly through the order book. The linear model treats each rung as an independent agent with constant marginal value; the log model captures a single agent expressing a concave demand curve.

Inventory risk penalizes concentration. An MM that buys 1,000 shares of Yes at \$0.50 has \$500 of inventory risk. A linear-utility MM is indifferent — the 1,001st share has the same marginal value as the 1st. In practice, every MM applies position limits: hard caps on per-market exposure. These are another piecewise-linear approximation of diminishing returns. Log utility makes position limits unnecessary — the natural concavity of ln penalizes concentration intrinsically.

4.4. Synthesis

Argument	**Why linear fails**	**Why log is right**
Kelly/Breiman	Bankrupt w.p. 1	Unique growth-optimal
Internal consistency	Budget + risk-neutral contradicts	Smooth risk aversion
Order ladders	Treats rungs as independent agents	Captures concave demand
Inventory risk	Requires bolted-on position limits	Concentration penalty intrinsic

The non-convexity of §3 is not a feature of prediction markets — it is an artifact of the linear welfare model.

5. The Main Result: Risk-Averse Clearing Is a Fisher Market

Replacing linear MM welfare with Kelly-criterion utility transforms the budget-constrained clearing problem into a convex program with a unique solution. Budget constraints disappear from the formulation — they are absorbed into the Eisenberg-Gale objective. The resulting program is structurally isomorphic to a quasi-linear Fisher market with endogenous supply.

5.1. The Program

Theorem 5 (Risk-Averse Clearing Is Convex). *Define the risk-averse batch auction clearing program with cash retention:*

$$P_b^{\mathrm{RA}}: \quad \max_{\boldsymbol{q}\in\mathcal{C},\boldsymbol{s}\geq 0} \quad \underbrace{\sum_k \Big[B_k \ln\Big(U_{k(\boldsymbol{q})} + s_k\Big) - s_k\Big]}_{\text{MM welfare}} + \underbrace{\sum_{j\notin \text{ MM}} w_j q_j}_{\text{retail welfare}} - \underbrace{C_{b(\boldsymbol{D}(\boldsymbol{q}))}}_{\text{minting cost}}$$

where MM_k^+ *is the set of MM k's buy orders (sell orders from MMs enter the retail welfare term; see §5.5),* $U_{k(\boldsymbol{q})} = \sum_{i\in \mathrm{MM}_k^+} L_i q_i \geq 0$ *is MM k's total weighted fill* ($L_i > 0$ *for all* $i \in \mathrm{MM}_k^+$), $s_k \geq 0$ *is MM k's retained cash,* $\mathcal{C} = \Big\{\boldsymbol{q} \in \Big[0, \overline{\boldsymbol{Q}}\Big] : \text{balance constraints}\Big\}$ *is the LP feasible set, and* C_b *is the smoothed minting cost (Proposition 1). Define* $\mu_k = B_k/(U_k + s_k)$ *as the shadow price of capital. Then:*

1. *The objective is strictly concave.* P_b^{RA} *has a unique optimum* $(\boldsymbol{q}^*, \boldsymbol{s}^*)$.
2. *Limit orders are exact:* $\mu_k \leq 1$, *so no MM order fills at negative welfare* ($L_i < p_k$).
3. *No explicit budget constraints appear. At the optimum, each MM k spends at most* B_k: *capital on fills plus retained cash* $\sum_{i\in \mathrm{MM}_k^+} p_{m(i)} q_i + s_k \leq B_k$.
4. *Clearing prices* $\boldsymbol{p}^*$ *are unique.*
5. *The program is solvable in polynomial time by any standard convex optimizer.*
6. *The program operates in two regimes per MM:*
   - *Capital-constrained* ($U_k \geq B_k$): $s_k = 0$, $\mu_k = B_k/U_k < 1$. *The MM prioritizes highest-ROI fills, organically staying within budget.*
   - *Over-capitalized* ($U_k < B_k$): $s_k = B_k - U_k$, $\mu_k = 1$. *The MM clears identically to the risk-neutral LP, absorbing excess budget into cash.*

5.2. Proof

(1) Strict concavity and uniqueness. Each $B_k \ln(U_k + s_k)$ is the composition of ln (strictly concave, increasing) with a positive affine function of $(\boldsymbol{q}, s_k)$ — strictly concave. The $-s_k$ term is linear. Retail welfare is linear. $-C_b$ is concave. The sum is strictly concave. The objective tends to $-\infty$ as $s_k \to \infty$ (since $-s_k$ dominates $\ln s_k$) and as $\boldsymbol{q}$ approaches the boundary of $\mathcal{C}$ from outside (box constraints). Therefore the superlevel sets $\big\{(\boldsymbol{q}, \boldsymbol{s}) : f^{\mathrm{RA}} \geq c\big\}$ are compact for any $c > -\infty$, so the supremum is attained. Strict concavity gives a unique maximizer. At the optimum, $U_k + s_k > 0$ for every MM with $B_k > 0$: the gradient $\frac{\partial}{\partial s_k}[B_k \ln(U_k + s_k) - s_k] \to +\infty$ as $U_k + s_k \to 0^+$, so the boundary $U_k + s_k = 0$ is never optimal.

(2) Limit order exactness. The KKT condition for $s_k \geq 0$ is:

$$\frac{\partial}{\partial s_k}: \quad \mu_k - 1 \leq 0, \quad s_k \geq 0, \quad (\mu_k - 1)s_k = 0$$

where $\mu_k = B_k/(U_k + s_k)$. This gives $\mu_k \leq 1$ unconditionally. The KKT for fill q_i of agent k is $\mu_k L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$. A fill ($q_i > 0$) requires $\mu_k L_i \geq p_{m(i)}$, hence $L_i \geq p_{m(i)}/\mu_k \geq p_{m(i)}$: the limit price must exceed the clearing price. No negative-welfare fill is possible.

The two regimes follow from complementary slackness: $s_k = 0$ when $\mu_k < 1$ (i.e., $U_k > B_k$), and $s_k = B_k - U_k > 0$ when $\mu_k = 1$ (i.e., $U_k < B_k$). In the over-capitalized regime ($\mu_k = 1$), the fill condition $L_i \geq p_{m(i)}$ is exactly the risk-neutral UCP condition — the MM clears identically to the LP.

(3) Budget absorption. Multiply the fill KKT by q_i and sum over $i \in \mathrm{MM}_k^+$. By complementary slackness ($\lambda_i^- q_i = 0$):

$$\mu_k \underbrace{\sum_{i \in \mathrm{MM}_k^+} L_i q_i}_{=U_k} = \sum_{i \in \mathrm{MM}_k^+} p_{m(i)} q_i + \underbrace{\sum_{i \in \mathrm{MM}_k^+} \lambda_i^+ q_i}_{\geq 0}$$

So $\sum p_{m(i)} q_i \leq \mu_k U_k$. Since $\mu_k(U_k + s_k) = B_k$, we have $\mu_k U_k = B_k - \mu_k s_k \leq B_k$:

$$\sum_{i \in \mathrm{MM}_k^+} p^*_{m(i)} q_i^* + s_k^* \leq B_k$$

Each MM's total deployment — capital on fills plus retained cash — is at most B_k. (The gap $\sum \lambda_i^+ q_i$ is infra-marginal surplus from fully-filled orders — profit that does not require additional capital.) The budget emerges from the objective, not from an explicit constraint. This is the Eisenberg-Gale mechanism extended to quasi-linear utilities: the ln singularity absorbs the budget, and the cash variable absorbs the surplus.

(4) Price uniqueness ($b > 0$). Prices are $p_k = \partial C_b / \partial D_k = \mathrm{softmax}(\boldsymbol{D}/b)$, a continuous function of the unique $\boldsymbol{q}^*$. (At $b = 0$, fills are unique but prices are LP duals, which may not be unique when multiple D_k tie at $\max_j D_j$; see §5.3.)

(5) Polynomial solvability. For $b > 0$, the objective is concave and C^∞. The feasible set is a polytope times $\mathbb{R}_+^{|\mathcal{K}|}$. Standard interior-point methods solve this in polynomial time. At $b = 0$, the objective is concave but non-smooth; subgradient or bundle methods apply. □

5.3. Temperature Independence

In the risk-neutral model, entropy smoothing was the only source of concavity — insufficient against budget non-convexity (Proposition 2). Here, $B_k \ln(U_k + s_k)$ provides b-independent curvature. Even at $b = 0$ (pure LP minting cost), the program remains strictly concave:

$$P_0^{\mathrm{RA}}: \quad \max_{\boldsymbol{q} \in \mathcal{C}, \boldsymbol{s} \geq 0} \quad \sum_k [B_k \ln(U_k + s_k) - s_k] + \sum_{j \notin \mathrm{MM}} w_j q_j - \max_k D_k$$

The $-\max_k D_k$ term is concave (not strictly), but the ln term provides strict concavity. Fill uniqueness, limit order exactness, and budget absorption hold at every temperature, including $b = 0$. Price uniqueness requires $b > 0$: at $b = 0$, prices are LP dual variables, which may not be unique when multiple outcomes k tie at $D_k = \max_j D_j$ (the subdifferential of max is a face of the simplex). In practice this is a non-issue: any $b > 0$ restores uniqueness, and the LMSR subsidy is bounded by $b \ln K$ (Proposition 1). This subsidy is funded by a permissionless liquidity pool; since batches are discrete, liquidity providers can deposit or withdraw between batches. The parameter b reflects pool depth — small b gives sharp prices with low subsidy. Setting b to a sub-penny value gives unique prices with negligible cost.

5.4. Welfare Convergence

The cash retention variable makes the risk-averse program a strict generalization of the risk-neutral LP: when budgets are non-binding, the two programs agree exactly.

> **Proposition 3** (LP Recovery). *Let $\boldsymbol{q}^{\mathrm{LP}}$ be the unconstrained LP optimum (P_b without budgets). If $B_k \geq U_k^{\mathrm{LP}} = \sum_{i \in \mathrm{MM}_k^+} L_i q_i^{\mathrm{LP}}$ for every MM k, then the risk-averse optimum $(\boldsymbol{q}^*, \boldsymbol{s}^*)$ satisfies $\boldsymbol{q}^* = \boldsymbol{q}^{\mathrm{LP}}$ and $s_k^* = B_k - U_k^{\mathrm{LP}}$. Fills, prices, and welfare are identical.*

Proof. When $B_k \geq U_k^{\mathrm{LP}}$ for all k, every MM is in the over-capitalized regime at $\boldsymbol{q}^{\mathrm{LP}}$: the optimizer sets $s_k = B_k - U_k^{\mathrm{LP}} \geq 0$ and $\mu_k = 1$. The fill KKT reduces to $L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$ — exactly the LP KKT. Since the LP and risk-averse programs have the same KKT conditions at $\boldsymbol{q}^{\mathrm{LP}}$, and the risk-averse program has a unique optimum, $\boldsymbol{q}^* = \boldsymbol{q}^{\mathrm{LP}}$. □

The cash variable s_k acts as a *numeraire good*: the MM can "buy" cash at price \$1, receiving no market exposure. This eliminates the over-fill pathology of the naive $B_k \ln U_k$ model (which forces all budget into fills, including unprofitable ones). The program interpolates smoothly between two regimes:

- *Capital-constrained* ($U_k > B_k,\ s_k = 0$): the MM has more profitable opportunities than budget. Kelly sizing kicks in — the MM prioritizes the highest return-on-investment fills, deploying the full budget. This is where the log model departs from the LP.
- *Over-capitalized* ($U_k < B_k,\ s_k > 0$): the MM has more budget than profitable fills. Excess capital is retained as cash ($\mu_k = 1$), and fills match the LP exactly. No order fills below its limit price. This is not a contradiction of Kelly — it is the Kelly prescription: bet a fraction of your bankroll proportional to your edge, keep the rest in cash. When the bankroll dwarfs the opportunity set, the Kelly fraction covers every profitable bet and the MM looks risk-neutral. The risk aversion is still there; it just doesn't bind.

Remark. Fractional Kelly sizing ($\alpha B_k \ln(U_k + s_k)$ for $\alpha < 1$) preserves strict concavity but increases the welfare gap: the effective budget αB_k is smaller, so more MMs become capital-constrained. The result is a reparametrization, not a structural extension.

Proposition 4 (Welfare Gap Bound). *Let W^{LP} and W^{RA} denote the total welfare (LP objective value) at the LP and risk-averse optima respectively, and let $\Delta_k = \max\big(0, U_k^{\mathrm{LP}} - B_k\big)$ be MM k's budget shortfall. Then:*

$$W^{\mathrm{LP}} - W^{\mathrm{RA}} \le \sum_k \left[\Delta_k - B_k \ln\left(1 + \frac{\Delta_k}{B_k}\right)\right] \le \sum_k \frac{\Delta_k^2}{2B_k}$$

Over-capitalized MMs ($\Delta_k = 0$) contribute nothing. The bound is quadratic in the shortfall.

Proof. For any $\boldsymbol{q}$ and cash choice $\boldsymbol{s}$ with $V_k = U_k + s_k$, the LP and RA objectives differ by $f^{\mathrm{LP}}(\boldsymbol{q}) - f^{\mathrm{RA}}(\boldsymbol{q}, \boldsymbol{s}) = \sum_k \varphi_k(V_k)$ where $\varphi_k(V) = V - B_k \ln V$ (this follows from $U_k = V_k - s_k$ and cancellation of s_k). Decompose:

$$W^{\mathrm{LP}} - W^{\mathrm{RA}} = \underbrace{\big[f^{\mathrm{LP}}(\boldsymbol{q}^{\mathrm{LP}}) - f^{\mathrm{RA}}(\boldsymbol{q}^{\mathrm{LP}}, \boldsymbol{s}^*)\big]}_{=\sum_k \varphi_k(V_k^{\mathrm{LP}})} + \underbrace{\big[f^{\mathrm{RA}}(\boldsymbol{q}^{\mathrm{LP}}, \boldsymbol{s}^*) - f^{\mathrm{RA}}(\boldsymbol{q}^{\mathrm{RA}}, \boldsymbol{s}^{\mathrm{RA}})\big]}_{\le 0} + \underbrace{\big[f^{\mathrm{RA}}(\boldsymbol{q}^{\mathrm{RA}}, \boldsymbol{s}^{\mathrm{RA}}) - f^{\mathrm{LP}}(\boldsymbol{q}^{\mathrm{RA}})\big]}_{=-\sum_k \varphi_k(V_k^{\mathrm{RA}})}$$

where $\boldsymbol{s}^*$ is the optimal cash at $\boldsymbol{q}^{\mathrm{LP}}$ and the middle term is non-positive by RA optimality. The function φ_k is convex with minimum $\varphi_k(B_k)$ at $V = B_k$. Since $\big(\boldsymbol{q}^{\mathrm{RA}}, \boldsymbol{s}^{\mathrm{RA}}\big)$ is the RA optimum, $\mu_k = B_k / V_k^{\mathrm{RA}} \le 1$ holds (part 2 of Theorem 5), giving $V_k^{\mathrm{RA}} \ge B_k$ and hence $\varphi_k\big(V_k^{\mathrm{RA}}\big) \ge \varphi_k(B_k)$:

$$W^{\mathrm{LP}} - W^{\mathrm{RA}} \le \sum_k \big[\varphi_k\big(V_k^{\mathrm{LP}}\big) - \varphi_k(B_k)\big]$$

At $\boldsymbol{q}^{\mathrm{LP}}$, the optimal cash is $s_k^* = \max\big(0, B_k - U_k^{\mathrm{LP}}\big)$, so $V_k^{\mathrm{LP}} = \max\big(U_k^{\mathrm{LP}}, B_k\big)$. Over-capitalized MMs have $V_k^{\mathrm{LP}} = B_k$ (zero contribution). Capital-constrained MMs contribute $\varphi_{k(U_k^{\mathrm{LP}})} - \varphi_{k(B_k)} = \Delta_k - B_k \ln(1 + \Delta_k / B_k)$. The quadratic bound follows from $\ln(1 + x) \ge x - x^2/2$. □

The quadratic bound is tight in the worst case: when all of MM k's orders are marginal ($L_i \approx p_k$), the RA solution achieves $V_k^{\mathrm{RA}} \approx B_k$ and the gap saturates the bound. In practice this is conservative — real order ladders have spread-out limit prices, so $V_k^{\mathrm{RA}} \gg B_k$ and the actual gap is much smaller. For moderate shortfalls ($\Delta_k / B_k \approx 0.5$), the exact bound gives $\approx 0.1 B_k$ versus the naive linear estimate $0.5 B_k$. For large shortfalls ($\Delta_k \gg B_k$), the exact expression grows as $\Delta_k - B_k \ln(\Delta_k / B_k)$ — sublinear in Δ_k, because the log model's prioritization becomes more efficient as the shortfall grows: the throttled fills are precisely the least profitable ones.

5.5. The Fisher Market Isomorphism

In a Fisher market (Eisenberg & Gale 1959), n consumers with budgets B_k purchase divisible goods with fixed supply s_j at prices p_j. The equilibrium is the unique solution to:

$$\mathrm{EG}: \quad \max_{\boldsymbol{x} \geq 0} \quad \sum_k B_k \ln U_k(\boldsymbol{x}_k) \quad \text{s.t.} \quad \sum_k x_{kj} \leq s_j \quad \forall j$$

where $U_{k(\boldsymbol{x}_k)} = \sum_j u_{kj} x_{kj}$ is consumer k's linear utility over goods. The supply constraints have dual variables p_j — the equilibrium prices. Budget constraints do not appear explicitly; they emerge from the $B_k \ln U_k$ objective by the same telescoping argument as our proof of (3). Adding a cash variable s_k with cost $-s_k$ yields the *quasi-linear* Fisher market (Cole et al. 2017, Chen et al. 2009) — agents can retain unspent budget as cash rather than being forced to spend it all on goods. Quasi-linear Fisher markets are well-studied: equilibrium existence, uniqueness, and polynomial-time computation are known under mild conditions.

Program P_b^{RA} is a quasi-linear Fisher market with two extensions:

Component	**Fisher market (EG)**	**Batch auction ($\boldsymbol{P_b^{\mathrm{RA}}}$)**
Consumers	n agents with budgets B_k	MMs with budgets B_k
Goods	Divisible commodities	Outcome shares
Supply	Fixed endowment s_j	Endogenous: minting at cost C_b
Utility	$U_k = \sum_j u_{kj} x_{kj}$	$U_k = \sum_{i \in \mathrm{MM}_k^+} L_i q_i$
Cash	Optional ($s_k \geq 0$)	Retained budget ($s_k \geq 0$)
Prices	Dual of $\sum_k x_{kj} \leq s_j$	Gradient of C_b (softmax)

The prediction market extends the quasi-linear Fisher market in two ways: (1) supply is endogenous — shares are created by minting at cost $C_{b(\boldsymbol{D})}$ rather than drawn from a fixed endowment, and (2) non-MM ("retail") orders contribute linear welfare alongside the log-utility MMs. Both extensions preserve concavity. The minting cost replaces the fixed supply constraints: where EG has $\sum_k x_{kj} \leq s_j$ with dual prices, P_b^{RA} has $C_{b(\boldsymbol{D})}$ whose gradient *is* the price vector. The cash variable s_k ensures that over-capitalized MMs park excess budget rather than distorting fills — the quasi-linear structure is essential, not optional.

5.6. What Changes and What Doesn't

What changes. Under P_b^{RA}, capital-constrained MMs ($\mu_k < 1$) prioritize their highest-ROI fills rather than filling every profitable order equally. An MM concentrating capital on one market faces the ln penalty: the first dollar of fill generates high marginal utility, additional dollars generate diminishing utility. This naturally diversifies MM capital across markets. Over-capitalized MMs ($\mu_k = 1$) behave identically to the LP — the log model only affects sizing when the budget actually binds.

What doesn't change. Non-MM orders are still matched by UCP at clearing prices. The minting mechanism is unchanged. Prices are still softmax (for $b > 0$) or LP duals (for $b = 0$). Limit orders are exact: no order fills below its stated price.

The buy/sell decomposition. The Fisher market isomorphism applies to MM buy orders — the orders that deploy capital and create demand. MM sell orders (if any) are treated as retail: filled by UCP with linear welfare, consuming no budget. This is economically correct: buying creates exposure and depletes the trading budget; selling liquidates existing inventory and *frees* budget. A sell fill returns capital to the MM rather than consuming it, so it should not appear inside $B_k \ln(U_k + s_k)$.

Remark on the decomposition. Under UCP, an MM's buy and sell on the same outcome never co-fill: a buy at limit L^{buy} fills when $L^{\text{buy}} \geq p_k$, a sell at limit L^{sell} fills when $p_k \geq L^{\text{sell}}$. For non-crossing orders ($L^{\text{sell}} > L^{\text{buy}}$), no clearing price p_k satisfies both; crossing orders ($L^{\text{sell}} \leq L^{\text{buy}}$) are self-dealing and rejected by the exchange at submission. Per outcome per batch, the MM is either buying (capital-consuming, enters ln) or selling (capital-freeing, enters retail welfare), never both.

Extension to multiple groups and bundle orders. Nothing in Theorem 5 requires a single mutually exclusive group. Consider N groups, each with K_j mutually exclusive outcomes. The joint state space is $\mathcal{S} = \prod_j \{1, ..., K_j\}$ (exactly one joint state is realized). Bundle orders — orders whose payoffs depend on the joint state — create demand D_s over $\mathcal{S}$. The minting cost generalizes to $V(\boldsymbol{D}) = \max_s D_s$ (minting one complete set of all joint states costs \$1), and the smoothed version is $C_b = b \ln \sum_s \exp(D_s/b)$. Both are convex in $\boldsymbol{D}$, and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$. The proof of Theorem 5 goes through unchanged for any fixed state space representation: $B_k \ln(U_k + s_k)$ is still strictly concave, $-C_b$ is still concave, budget absorption and limit order exactness still hold. The open question is whether the program can be solved without explicitly enumerating the joint states (see §6.2).

When no orders span multiple groups, the joint problem decomposes: $C_b = \sum_j C_b^{G_j}$ (the product-LMSR factorization of Chen and Pennock 2007). Cross-group orders break this separability, coupling groups transitively — if orders link groups A–B and B–C, the clearing problem requires the full $K_A \times K_B \times K_C$ state space. The mathematical obstruction is purely computational, not structural (see §6.2).

6. Discussion

6.1. Summary

The paper establishes one main result and the framework necessary to state and prove it:

Section	Result	Role
§2	LMSR = smoothed LP (Thms 1–4)	Framework
§3	Computational obstruction (Prop. 2)	Motivation
§4	Log utility is correct model	Economic argument
§5	Fisher market isomorphism (Thm 5); LP recovery (Prop. 3); welfare gap (Prop. 4)	**Main result**

The landscape: budget-constrained risk-neutral clearing is a non-convex bilinear program with no known polynomial-time algorithm (Proposition 2); risk-averse clearing is a standard convex program with guaranteed uniqueness and exact limit orders (Theorem 5). When budgets are non-binding, the two programs agree exactly (Proposition 3). When they bind, the welfare gap is quadratic in the budget shortfall (Proposition 4). The transition from intractability to tractability requires only the change of objective from $\sum w_i q_i$ to $\sum B_k[\ln(U_k + s_k) - s_k]$.

6.2. Open Problems

1. **Welfare gap in practice.** Proposition 4 bounds the welfare gap by $\sum \Delta_k^2/(2B_k)$. For typical order books, is the bound tight? Empirical comparison between P_b^{RA} and

the LP across realistic MM ladder structures would quantify the practical cost of risk-averse clearing.

2. **Efficient combinatorial clearing.** The Fisher market isomorphism extends to joint state spaces (§5.5), but the state space $|\mathcal{S}| = \prod K_j$ is exponentially large. Cross-group bundle orders couple groups transitively: if orders span A–B and B–C, clearing requires the full $K_A \times K_B \times K_C$ space. In practice, a handful of bundle orders can connect all groups. Each order's payoff is a low-rank tensor (touching ≤ 5 groups), so the demand D_s has exploitable structure. Can the EG program be solved in time polynomial in the number of orders rather than the state space? The analogous question for combinatorial LMSR (without budgets) is already open; budgets add a further layer.

3. **Risk-neutral hidden convexity.** Is the budget-constrained risk-neutral problem actually convex despite the bilinear constraints? The LMSR softmax prices grow fast enough that concentrated positions are budget-expensive, and we have found no instance with non-unique clearing prices. A proof of uniqueness (or a counterexample) would settle whether the computational obstruction of §3 is fundamental or merely an artifact of current proof techniques. Devanur and Dudík (2015) proved budget additivity for sequential LMSR, hinting at hidden convexity — but extending this to simultaneous batch auctions remains open.

6.3. Connection to Prior Work

Eisenberg and Gale (1959): Convex program for Fisher market equilibrium via expenditure variables. Our Theorem 5 establishes that prediction markets with risk-averse MMs are isomorphic to Fisher markets: the Eisenberg-Gale program with endogenous supply (minting cost C_b) is convex. §3 identifies why the same approach fails for risk-neutral MMs: constant marginal returns provide no curvature to counteract the budget non-convexity.

Abernethy, Chen, and Vaughan (2013): Proved that cost-function market makers satisfying a set of axioms must price via a convex cost function, with LMSR corresponding to the entropy conjugate. Our §2 is the batch-auction formulation of their framework. The minting cost $V = \max_k D_k$ is the "simplest" cost function, with LMSR as its entropy smoothing.

Breiman (1961): Optimal properties of the Kelly criterion. Our §4 uses Breiman's survival theorem to argue that log utility is not a convenience but a consequence of evolutionary selection among repeat-participation MMs.

Devanur and Dudík (2015): Price uniqueness and budget additivity for sequential LMSR via Bregman divergence. Their budget additivity is a manifestation of hidden convexity — the bilinear budget boundaries merge into a convex hull in the dual space. Our Theorem 5 achieves unconditional uniqueness for batch auctions via the same Eisenberg-Gale structure underlying their result. Whether the *risk-neutral* batch auction also has hidden convexity remains open (§3 shows it cannot come from the standard Fenchel dual).

Fortnow, Kilian, Pennock, and Wellman (2005): LP for combinatorial call markets. Our group minting provides the structural reason why LP works: it encodes mutual exclusivity in $O(K)$ vs $O(2^K)$.

Chen and Pennock (2007): Bounded-loss market makers. The parameter b in our framework is their loss bound. $b = 0$ (zero loss) is achievable in the batch setting by complementary slackness: the minting mechanism breaks even exactly when prices are LP duals.

Hofbauer and Sandholm (2007): Unique logit equilibrium for negative semidefinite games — games where increasing adoption of a strategy decreases its payoff. Our prediction market is exactly NSD (demand raises price). Their result covers the unconstrained case; our Fisher market isomorphism extends to budget-constrained agents unconditionally.

McKelvey and Palfrey (1995): Quantal Response Equilibrium. The budget-constrained clearing problem is the prediction-market analog of QRE with budget constraints. Our impossibility result (Proposition 2) corresponds to the failure of high-temperature contraction when budgets bind; our resolution via log utility corresponds to adopting a different utility model entirely.

Cole et al. (2017) and Chen et al. (2009): Quasi-linear Fisher markets with spending constraints. Our P_b^{RA} is structurally a quasi-linear Fisher market; the contribution is not the quasi-linear framework itself but its application to prediction markets with endogenous supply (minting cost C_b), mixed agent types (log-utility MMs alongside linear-welfare retail), and the connection to LMSR cost-function market makers.

Budish, Cramton, and Shim (2015): Frequent Batch Auctions for equity markets. Our framework applies FBAs to prediction markets, where the information-driven price shocks are even more severe than in equities.

Next steps: (1) Implement risk-averse clearing (Theorem 5) — single convex program, no annealing. (2) Empirical welfare comparison between P_b^{RA} and risk-neutral LP across realistic order books, measuring the welfare gap of Proposition 4 for typical MM ladder structures.